INTERNATIONAL SOCIETY FOR PHARMACOEPIDEMIOLOGY

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Let’s Get Personal

Defining and Estimating Effects of Dynamic Treatment Strategies Using Real-World Data

Lecture 2

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Outline

- **Estimating the effects of dynamic treatment strategies**
  IP weighting of a dynamic marginal structural model

- **Key assumptions**

- **Data requirements**

- **Alternative estimators**

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Estimation procedure

- We focus on one of the most common approaches for per-protocol effects
  - **Inverse-probability (IP) weighting** of a **dynamic marginal structural model**

- The procedure can be summarized in **3 (simple?) steps**
  - *Step 1*: Construct IP weights
  - *Step 2*: Apply censoring rules
  - *Step 3*: Estimate effects via a dynamic marginal structural model

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Step 1: Construct IP weights

- IP weighting **accounts for confounding**
  - i.e., non-random non-adherence to the dynamic treatment strategies

Step 1: Construct IP weights

- Recall: Non-stabilized weights W^A for a **time-fixed treatment** A

$$W^A = \frac{1}{f(A|L)} \qquad If\ treated: \quad W^A = \frac{1}{Pr(A=1|L)}$$

$$If\ untreated: \quad W^A = \frac{1}{Pr(A=0|L)}$$

- Denominator is (informally) the probability of having your observed treatment value A given your confounder values L

Step 1: Construct IP weights

- But we are interested in sustained, dynamic strategies
  - where treatment can vary over time
  - which means confounders can also vary over time
- We need: IP weights for a **time-varying treatment** $\bar{A}$ at time k

$$W_k^{\bar{A}} = \frac{1}{f(A_0|\bar{A}_{-1}, \bar{L}_0)} \times \frac{1}{f(A_1|\bar{A}_0, \bar{L}_1)} \times \cdots \frac{1}{f(A_k|\bar{A}_{k-1}, \bar{L}_k)} = \prod_{m=0}^{k} \frac{1}{f(A_m|\bar{A}_{m-1}, \bar{L}_m)}$$

Time-specific contribution

Final weight at time k

- *Overbars denote history of a variable*
- *Subscripts represent time points*

Step 1: Construct IP weights

- To estimate the weights

$$W_k^{\bar{A}} = \prod_{m=0}^{k} \frac{1}{f(A_m|\bar{A}_{m-1}, \bar{L}_m)}$$

- Fit **model(s) for the denominator probabilities**
  - for example, for a binary treatment, we may fit:

$$\text{logit}[\Pr(A_m = 1|\bar{A}_{m-1}, \bar{L}_m)]$$

Step 1: Construct IP weights

- But for dynamic strategies, **special considerations** are needed
  - because treatment decisions depend on evolving patient characteristics

Step 1: Construct IP weights

- **<u>Consideration #1</u>:** For strategies that include **excused** times (when treatment decisions are not enforced but up to clinical discretion), weight construction needs to be modified
  - At excused times:
    - the **time-specific contribution** to the weights is **1**
    - observations are generally **excluded from the model** used to estimate denominator probabilities

Medication class can be changed if severe urinary tract infection, diabetic ketoacidosis, pancreatitis, or medullary thyroid cancer develops.

Excused

Step 1: Construct IP weights

- **<u>Consideration #2</u>:** The usual stabilized weights **cannot be used**
  - Because the numerator of these weights depends on an individual's *time-varying treatment*
    - See Technical Point 21.2 in Hernán MA, Robins JM (2020). *Causal Inference: What If*.
  - Alternative stabilization approaches are available (e.g., based only on baseline variables, see Cain et al. Int J Biostat. 2010)
    - but they are *not guaranteed to produce more precise estimates*

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Step 2: Apply censoring rules

- Censoring ensures all individuals **follow their assigned strategy**
  - Individuals are censored due to non-adherence whenever they **deviate** from their assigned treatment strategy

Step 2: Apply censoring rules

- Censoring rules are a function of the treatment strategies (and vice versa)
  - **deviations** from the strategy are handled via censoring
  - if **censoring** is used, the analysis **generally implies an intervention** under which the condition that triggered censoring is not allowed

Step 2: Apply censoring rules

- For dynamic strategies, **special considerations** are needed
  - because treatment decisions (and therefore censoring rules) depend on evolving patient characteristics

Step 2: Apply censoring rules

- **<u>Consideration #1</u>:** Non-adherence to the strategies (deviations) can be a function of not just treatment A but also covariates L
  - individuals are censored when they deviate

Deviated (censored)

Any contraindication? No!

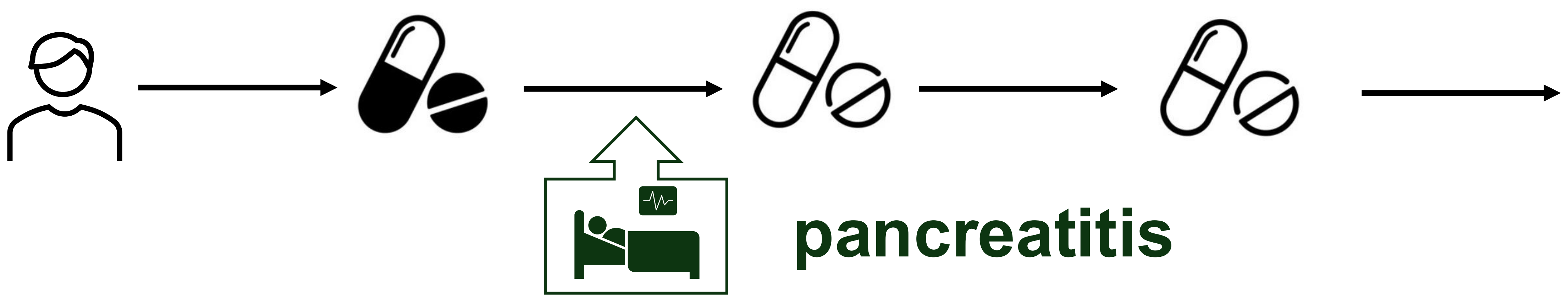
Discontinuation of treatment *prior to contraindication* is not allowed

Deviation

Step 2: Apply censoring rules

- <u>Consideration #2</u>: For strategies that include **excused** conditions, individuals cannot be censored if and when they are excused
  - Careful strategy specification is essential ***before*** censoring

Excused (NOT censored)



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Step 3: Estimate effects

- To estimate effects, fit an outcome model to the IP weighted and censored data

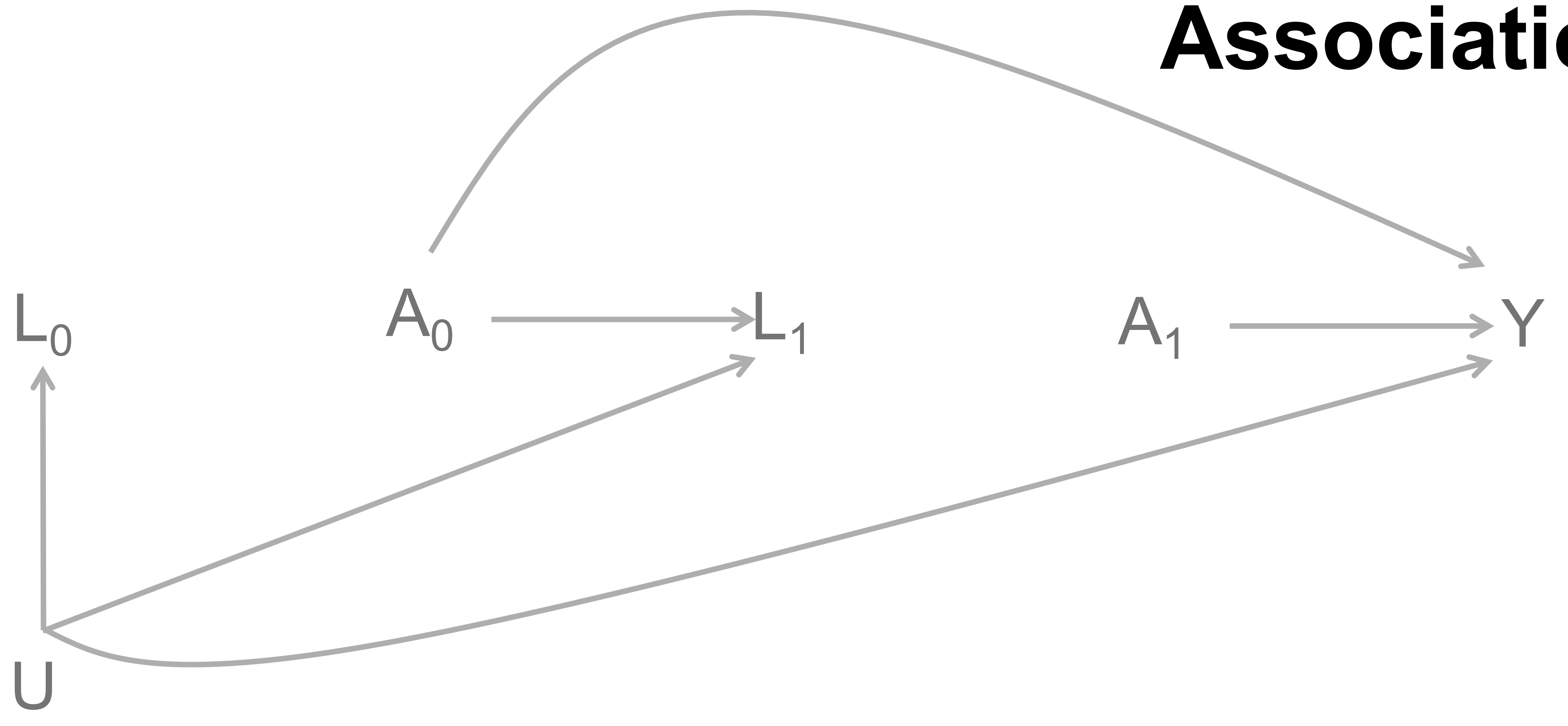
- Parameters from this model estimate parameters of a **dynamic marginal structural model (MSM)**, a *causal* model for *counterfactual outcomes*
  - Why?
  - Because in the IP weighted population, if assumptions hold

    association (weighted outcome model) = causation (MSM)

Step 3: Estimate effects

- Weights create a pseudo-population in which treatment is independent of measured, time-varying confounders

Association = causation



Pseudo-population for the non-stabilized weights

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Key assumptions

- We said that, in the pseudo-population, **if assumptions hold**

  association (weighted outcome model) = causation (MSM)

- That is, endowing estimates with a causal interpretation requires **assumptions**, typically the following identifying assumptions:
  - Sequential exchangeability
  - Sequential positivity
  - Sequential consistency

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Key assumptions

- **Sequential exchangeability***

$$Y^g \perp\!\!\!\perp A_k | \bar{A}_{k-1} = g(\bar{A}_{k-2}, \bar{L}_{k-1}), \bar{L}_k \text{ for all strategies } g \text{ and } k = 0, 1, \ldots K$$

  - the counterfactual outcome under dynamic strategy g, Y^g, is independent of treatment at every time given *past* treatment and covariates compatible with g
    - i.e., no unmeasured time-fixed or time-varying confounding at any time
    - but if an individual is ***excused***, then this assumption is no longer required!

Medication class can be changed if severe urinary tract infection, diabetic ketoacidosis, pancreatitis, or medullary thyroid cancer develops.

Excused

* For different versions of sequential exchangeability (including weaker versions that can be used to identify static strategies and stronger versions that are required when treatment strategies depend on the natural value of treatment), see *Causal Inference: What If?* Section 19.5 and Technical Point 19.3 and Sarvet and Stensrud. 2026. *Annual Review of Statistics and Its Application.*

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Key assumptions

- **Sequential positivity**

If the joint probability of a treatment and covariate history $(\bar{a}_{k-1}, \bar{l}_k)$ at time k is nonzero, then the conditional probability of receiving each type of treatment (a_k) at time k must be nonzero

  - i.e., everyone has a chance to receive all levels of treatment at each time within levels of past treatment and confounders
  - but if an individual is **excused**, then this assumption is no longer required!

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Key assumptions

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Key assumptions

- **Sequential consistency***

For any strategy g, if $A_k = g_k(\bar{A}_{k-1}, \bar{L}_k)$ at each time k for a given individual then $Y^g = Y$

  - i.e., treatment is well-defined (only one version of treatment)
  - and there is no interference (treatment status of one individual does not affect the potential outcomes of other individuals)

*Stronger (generally accepted) versions of this assumption are discussed in *Causal Inference: What If?* Technical Point 19.2

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Data requirements

- What data do we need?
  - In short: we need good measures of ***every element*** **included in the target trial protocol**
- This includes everything needed to define
  - eligibility criteria
  - dynamic treatment strategies (*‘assignment,’ deviations, excusing*)
  - outcomes
  - follow-up
  - covariates (generally, *all time-fixed and time-varying confounders*)

Data requirements

- **Consideration**: for dynamic treatment strategies
  - determining whether a treatment change (e.g., discontinuation) is a deviation or consistent with the strategy generally requires
    - sufficiently detailed, time-updated data on *treatment changes* and *reasons* for treatment changes

  - Therefore, our treatment strategies often also require **frequent measurement** (or *opportunities for measurement*) of these variables

Individuals must also have a healthcare contact at least once every 12 months

Deviation

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Alternative estimators

- Estimating effects of dynamic strategies (*almost always*) requires **g-methods**
  - that is, methods designed to handle *time-varying confounding* and *treatment-confounder feedback**

* See *Causal Inference: What If?* Chapter 20 and Section 19.3

Alternative estimators

- We reviewed one g-method (**IP weighting of an MSM**)
    - but this is *not the only option*

- Other g-methods offer trade-offs in **statistical efficiency, modeling assumptions,** and **robustness**

Alternative estimators

- **Plug-in g-formula** (Robins 1986)
- **G-estimation of structural nested models** (Robins 1989, 1991)
- **Inverse probability weighting** (Robins 1998)
  - of a dynamic marginal structural model
  - of a dose-response marginal structural model (only sometimes possible, see Toh et al. 2012 for special considerations)
- **Doubly-robust versions** (Robins, van der Laan, Rotnitzky)
  - e.g., collaborative targeted minimum loss-based estimation, AIPW

Alternative estimators

- Methods can be extended and **combined** with other causal inference tools
    - sequential trial emulation
    - cloning
    - IP weighting for loss to follow-up
    - causal approaches for handling competing events and grace periods
    - optimal regimes
    - … etc.

- Choice of method(s) should be guided by the **question** and **available data**

Summary

- Estimating the effects of dynamic treatment strategies generally requires **g-methods**
  - such as IP weighting of a dynamic marginal structural model or other g-methods
- When using IP weighting of an MSM
  - weights and censoring rules need to be *tailored to* the dynamic strategy of interest
- Careful attention to **the validity of identifying assumptions** is essential
  - but dynamic strategies can also be used to *make some assumptions more plausible*!
- To estimate effects of dynamic treatment strategies, good methods aren't good enough . . . **high-quality longitudinal data** are also essential

Thank you!

Questions?

Supplemental

- Why are MSMs **“structural?”**
  - Because they are models for counterfactual outcomes, Y^g, which are referred to as “structural” or “causal” models, e.g.,

$$E[Y^g] = \beta_0 + \beta_1 g$$

  - Parameters for treatment in this structural model (β_1) have a direct causal interpretation

Supplemental

- Why are MSMs **“marginal?”**
  - Because they are models for functionals of the **marginal distribution** of the counterfactual outcome
  - They do not impose any restrictions on the joint distribution of counterfactual outcomes under different treatment strategies

model the mean of $\mathrm{Y}^{g=g1}$ and the mean of $\mathrm{Y}^{g=g2}$

but say nothing about the relationship between them